UNIVERSITY OF BIRMINGHAM

School of Physics and Astronomy

DEGREE OF B.Sc. & M.Sci. WITH HONOURS

THIRD-YEAR FINAL EXAMINATION

03 21826

LH THE LIFE AND DEATH OF STARS

SUMMER EXAMINATIONS 2025

Time allowed: 1 hour 30 minutes

Answer ***two*** questions. If you answer more than two questions, credit will only be given for the best two answers.

The approximate allocation of marks to each part of a question is shown in brackets **[ ]**.

Calculators may be used in this examination but must not be used to store text. Calculators with the ability to store text should have their memories deleted prior to the start of the examination.

A formula sheet and two tables of physical constants and units that may be required will be found at the end of this question paper.

Answer **two** *questions. If you answer more than two questions, credit will only be given for the best two answers.*

1. The figure below shows a Hertzsprung-Russell diagram of a stellar population, with 4 stars (A, B, C, D) indicated.

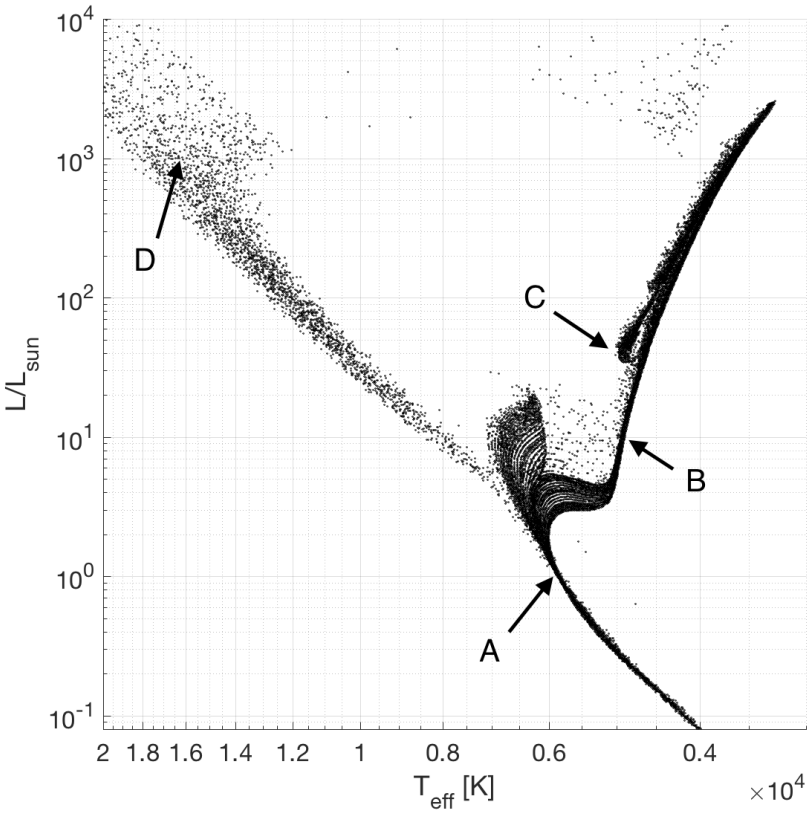
   (a) Which nuclear fuel is burnt in stars A, B, C, D, and where in the star does this occur? **[6]**

   (b) Do the stars shown in the HR diagram below all belong to the same stellar cluster? Explain the reasons for your answer clearly. **[5]**

   (c) Using the information in the figure, estimate the ratio of the radii of stars B and C, and the ratio of the radii of stars A and D. **[6]**

   (d) Estimate the ratio of mean densities of stars A and D. Clearly state your assumptions. **[8]**

   (e) Estimate the ratio of the dynamical time scales of stars A and D. Clearly state your assumptions. **[5]**



2. (a) Derive the equation for hydrostatic equilibrium, naming all variables and stating all assumptions used. Include a relevant sketch to illustrate your derivation. **[10]**

(b) The speed of sound, c_s, in a star is given by $c_s = \sqrt{\Gamma_1 P/\rho}$, where Γ_1 is the adiabatic exponent, P is the pressure, and ρ the density. Suppose a star has the mass and radius of the Sun and a constant density. Using the equation of hydrostatic equilibrium, compute the sound speed as a function of radius. **[7]**

(c) The time taken by a sound wave to travel radially from one radius r_1 to another radius r_2 is

$$t = \int_{r_1}^{r_2} \frac{dr}{c_s(r)}.$$

Show that the time taken for a sound wave to travel from the centre to the surface of the uniform star mentioned in (b) is

$$t = \sqrt{\frac{\pi^2}{2\Gamma_1}\frac{R^3}{GM}}.$$

Compute its value, in minutes, for $R = \mathrm{R}_\odot$, $M = \mathrm{M}_\odot$ and $\Gamma_1 = 5/3$. You will need the integral

$$\int_0^{r'} \frac{dr}{\sqrt{R^2 - r^2}} = \arcsin\frac{r'}{R}.$$

[6]

(d) Consider two diffferent stars, each with a core of fully ionised ideal gas. One star has a core with typical solar chemical composition while the other star's core consists purely of helium. How do the sound speeds in the cores of these stars compare? Clearly state all your assumptions. **[7]**

3. (a) By considering the radiation transfer across a thin shell, derive an approximation to the radiative transfer equation. Indicate how your derived equation differs from the actual equation (given in the Formula Sheet). Make a sketch and clearly state all your assumptions.

[Hint: The flux absorbed by a small shell is $dF = -\kappa\rho F\, dr$] **[10]**

(b) Assume that the central density of a 10 $\mathrm{M}_\odot$, 5 $\mathrm{R}_\odot$ star is 10 $\mathrm{g/cm^3}$ and that the opacity in the core is 0.5 $\mathrm{cm^2/g}$. Estimate the mean free path of a photon in the star's core. If this mean free path remained constant for the photon's journey to the surface, calculate the average time it would take for a photon to traverse the star. Compare this time with the nuclear timescale, and with the time for a photon to travel from the centre of the star to the surface if it did not interact at all with matter.

[Hint: use the fact that the net distance travelled by a photon that random walks through N steps of average length equal to the mean free path l is $d = l\sqrt{N}$.] **[10]**

(c) Describe the evolution in time of the hydrogen core mass fraction during the main-sequence in the following cases:

i. a 1 $\mathrm{M}_\odot$ star,
ii. a 5 $\mathrm{M}_\odot$ star.

For each case make a labeled sketch of the hydrogen mass fraction as a function of stellar mass, and provide arguments justifying your sketch. Clearly indicate the appropriate stellar ages for each star in your sketch. **[10]**

Formula Sheet

Equations of stellar structure

$$\text{Mass equation:}\quad \frac{dm}{dr} = 4\pi r^2 \rho$$

$$\text{Hydrostatic equilibrium:}\quad \frac{dP}{dr} = -\frac{G\,m\,\rho}{r^2}$$

$$\text{Energy equation:}\quad \frac{dL}{dr} = 4\pi\, r^2 \left[\rho\epsilon - \rho\frac{d}{dt}\left(\frac{u}{\rho}\right) + \frac{P}{\rho}\frac{d\rho}{dt}\right]$$

$$\text{Radiative energy transport equation:}\quad \frac{dT}{dr} = -\frac{3\,\kappa\rho\,L}{16\pi\,a\,c\,r^2 T^3}$$

$$\text{Convective energy transport equation:}\quad \frac{dT}{dr} = \frac{\Gamma_2 - 1}{\Gamma_2}\frac{T}{P}\frac{dP}{dr}$$

Ideal gas law

$$\rho = \frac{\mu m_u P}{k\,T}$$

Mean molecular weight

$$\mu^{-1} = \sum_j X_j \frac{Z_j + 1}{A_j}$$

Adiabatic exponents

$$\Gamma_1 = \left(\frac{\partial \ln P}{\partial \ln \rho}\right)_s$$

$$\frac{\Gamma_2}{\Gamma_2 - 1} = \left(\frac{\partial \ln P}{\partial \ln T}\right)_s$$

$$\Gamma_3 - 1 = \left(\frac{\partial \ln T}{\partial \ln \rho}\right)_s$$

Radiation constant

$$a = \frac{4\sigma}{c}$$

Radiation pressure

$$P_{\text{rad}} = \frac{1}{3} a\, T^4$$

Degenerate matter pressure

$$P_{\text{deg,nr}} \propto \rho^{5/3} \qquad\qquad P_{\text{deg,r}} \propto \rho^{4/3}$$

Stefan–Boltzmann law

$$L = 4\pi R^2 \sigma T_{\text{eff}}^4$$

$$F = \sigma T_{\text{eff}}^4$$

Lane-Emden equation

$$\frac{1}{\xi^2}\frac{d}{d\xi}\left(\xi^2\frac{d\theta}{d\xi}\right) = -\theta^n$$

Photon mean free path

$$\lambda = \frac{1}{\kappa_\nu\,\rho} = \frac{1}{n\,\sigma_R}$$

Convective instability criterion

$$\nabla_R = \frac{d\ln T}{d\ln P} > \nabla_{\text{ad}} = \frac{\Gamma_2 - 1}{\Gamma_2}$$

Coulomb energy

$$E_C = \frac{q_1\,q_2}{4\,\pi\,\epsilon_0\,r}$$

Thermal kinetic energy

$$E_{th} = \frac{3}{2}\,k\,T$$

Cross section and Gamow energy

$$\sigma(E) = \frac{S(E)}{E}\exp\left(-\sqrt{\frac{E_G}{E}}\right) \qquad \sqrt{E_G} = \frac{2\pi\,e^2\sqrt{m_u}}{\sqrt{2}\hbar\,4\pi\,\epsilon_0}Z_1\,Z_2\,\sqrt{A}$$

Released energy for reaction $A(a,y)Y$

$$Q = [\Delta m(A) + \Delta m(a) - \Delta m(Y) - \Delta m(y)]\,c^2$$

Average nuclear reaction rate is proportional to

$$\langle\sigma\,v\rangle = \left(\frac{8}{m\pi}\right)^{1/2}\frac{1}{(k\,T)^{3/2}}\int_0^\infty S(E)\exp\left(-\frac{E}{k\,T} - \frac{b}{\sqrt{E}}\right)dE$$

Maximum of the Gamow peak

$$E_0 = \left(\frac{b\,k\,T}{2}\right)^{2/3}$$

Astrophysical Constants and Units

Astronomical Unit	AU	1.50×10^{11} m
Parsec	pc	3.1×10^{16} m
Tropical year	y	365.242 mean solar days
Solar luminosity	$L_\odot$	3.84×10^{26} W
Absolute bolometric magnitude of the Sun	$M_{\rm bol}$	$+4^{\rm m}75$
Bolometric correction for the Sun	BC	$-0^{\rm m}08$
Apparent visual magnitude of the Sun	$m_v(\odot)$	$-26^{\rm m}74$
Solar constant	f	1.36×10^3 W m^{-2}
Solar mass	$M_\odot$	1.989×10^{30} kg
Solar radius	$R_\odot$	6.95×10^8 m
Solar effective temperature	$T_{\rm eff}$	5777 K
Wien constant	b	2.898×10^{-3} m K
Hubble constant	H_o	70 km s^{-1} Mpc^{-1}
Distance of the Sun from galactic centre		8.3 kpc
Earth mass	$M_\oplus$	5.972×10^{24} kg
Earth radius	$R_\oplus$	6371 km
Jupiter mass	$M_{\rm Jup}$	1.898×10^{27} kg
Jupiter radius	$R_{\rm Jup}$	71 492 km

Physical Constants and Units

Acceleration due to gravity	g	$9.81\,\mathrm{m\,s^{-2}}$
Gravitational constant	G	$6.674 \times 10^{-11}\,\mathrm{N\,m^2\,kg^{-2}}$
Ice point	T_{ice}	$273.15\,\mathrm{K}$
Avogadro constant	N_A	$6.022 \times 10^{23}\,\mathrm{mol^{-1}}$
		[*N.B.* 1 mole $\equiv 1$ *gram-molecule*]
Gas constant	R	$8.314\,\mathrm{J\,K^{-1}\,mol^{-1}}$
Boltzmann constant	k, k_B	$1.381 \times 10^{-23}\,\mathrm{J\,K^{-1}} \equiv 8.62 \times 10^{-5}\,\mathrm{eV\,K^{-1}}$
Stefan constant	σ	$5.670 \times 10^{-8}\,\mathrm{W\,m^{-2}\,K^{-4}}$
Rydberg constant	R_∞	$1.097 \times 10^{7}\,\mathrm{m^{-1}}$
	$R_\infty hc$	$13.606\,\mathrm{eV}$
Planck constant	h	$6.626 \times 10^{-34}\,\mathrm{J\,s} \equiv 4.136 \times 10^{-15}\,\mathrm{eV\,s}$
$h/2\pi$	$\hbar$	$1.055 \times 10^{-34}\,\mathrm{J\,s} \equiv 6.582 \times 10^{-16}\,\mathrm{eV\,s}$
Speed of light *in vacuo*	c	$2.998 \times 10^{8}\,\mathrm{m\,s^{-1}}$
	$\hbar c$	$197.3\,\mathrm{MeV\,fm}$
Charge of proton	e	$1.602 \times 10^{-19}\,\mathrm{C}$
Mass of electron	m_e	$9.109 \times 10^{-31}\,\mathrm{kg}$
Rest energy of electron		$0.511\,\mathrm{MeV}$
Mass of proton	m_p	$1.673 \times 10^{-27}\,\mathrm{kg}$
Rest energy of proton		$938.3\,\mathrm{MeV}$
One atomic mass unit	u	$1.66 \times 10^{-27}\,\mathrm{kg}$
Atomic mass unit energy equivalent		$931.5\,\mathrm{MeV}$
Electric constant	ϵ_0	$8.854 \times 10^{-12}\,\mathrm{F\,m^{-1}}$
Magnetic constant	μ_0	$4\pi \times 10^{-7}\,\mathrm{H\,m^{-1}}$
Bohr magneton	μ_B	$9.274 \times 10^{-24}\,\mathrm{A\,m^2}$ ($\mathrm{J\,T^{-1}}$)
Nuclear magneton	μ_N	$5.051 \times 10^{-27}\,\mathrm{A\,m^2}$ ($\mathrm{J\,T^{-1}}$)
Fine-structure constant	$\alpha = e^2/4\pi\epsilon_0\hbar c$	7.297×10^{-3}= 1/137.0
Compton wavelength of electron	$\lambda_c = h/m_e c$	$2.426 \times 10^{-12}\,\mathrm{m}$
Bohr radius	a_0	$5.2918 \times 10^{-11}\,\mathrm{m}$
angstrom	Å	$10^{-10}\,\mathrm{m}$
barn	b	$10^{-28}\,\mathrm{m^2}$
torr (mm Hg at $0\,^\circ$C)	torr	$133.32\,\mathrm{Pa}$ ($\mathrm{N\,m^{-2}}$)

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Do not complete the attendance slip, fill in the front of the answer book or turn over the question paper until you are told to do so.

Important Reminders

- Coats/outwear should be placed in the designated area.
- Unauthorised materials (e.g. notes or Tippex) must be placed in the designated area.
- Check that you do not have any unauthorised materials with you (e.g. in your pockets, pencil case).
- Mobile phones and smart watches **must** be switched off and placed in the designated area or under your desk. They must not be left on your person or in your pockets.
- You are not permitted to use a mobile phone as a clock. If you have difficulty seeing a clock, please alert an Invigilator.
- You are not permitted to have writing on your hand, arm or other body part.
- Check that you do not have writing on your hand, arm or other body part – if you do, you must inform an Invigilator immediately
- Alert an Invigilator immediately if you find any unauthorised item upon you during the examination.

Any students found with non-permitted items upon their person during the examination, or who fail to comply with Examination rules may be subject to Student Conduct procedures.